

Equipment Inspection Report

Document No: INSP-000015 | Refinery Operations — Confidential Internal Document

Equipment Tag	CL-102	Equipment Type	Distillation column
Zone	Crude distillation unit	Install Date	02 Mar 2012
Inspection Date	04 Mar 2023	Next Due Date	31 Aug 2023
Inspected By	Rabin Basnet (Maintenance Technician)	Result	FLAG

1. Scope of Inspection

Routine 180-day mechanical and process inspection of column CL-102 (crude distillation unit), covering tray condition, internals, feed nozzle integrity, and pressure boundary checks, per the unit's standard inspection schedule.

2. Findings

Visual and differential-pressure indications consistent with early-stage tray flooding were observed in the mid-column section. Tray fouling was noted on inspection trays 14-18, consistent with this equipment's known failure modes (tray fouling, flooding, corrosion at feed nozzle). No active leakage or structural damage was found at this time.

3. Assessment

Column CL-102 was installed on 02 Mar 2012 and carries an 18-year expected service life; it remains within its rated life at the time of this inspection. The flooding indication is assessed as an early operational finding rather than an end-of-life failure, and is recommended for corrective tray cleaning at the next planned shutdown window rather than immediate isolation.

4. Recommended Action

Schedule corrective maintenance for tray cleaning on CL-102 within the current maintenance cycle. Increase differential-pressure monitoring frequency on this column until the next inspection. Re-inspect at the standard 180-day interval; earlier re-inspection is warranted if differential pressure trends continue to rise.

Prepared by	Rabin Basnet	Signature	
Reviewed by	Sunil Maharjan (Reliability Engineer)	Date	04 Mar 2023